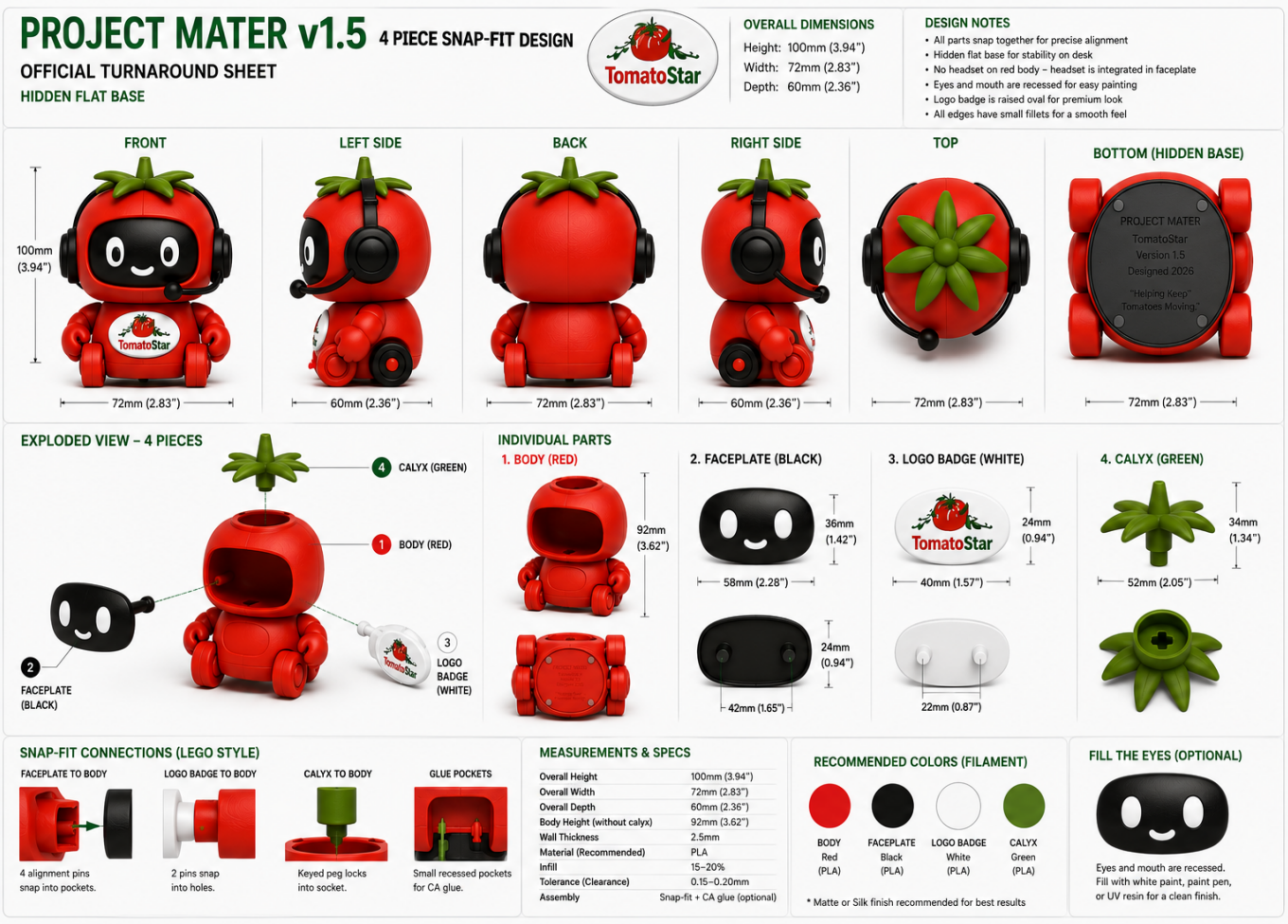
PROJECT MATER v1.5

Four-piece CAD engineering reference package

Frozen concept: red body with hidden flat base, black removable faceplate, white logo badge, and green Morning Star-style calyx. The faceplate contains the complete headset. No headset geometry is molded into the red head.



Important: the turnaround above is a visual concept reference. Build the CAD from the nominal dimensions and constraints in this brief, then validate all fits on the actual R&D; printer.

1. Deliverable Architecture

CAD component	Color	Included geometry	Connection
01_Body_Red	Tomato red	Tomato shell, torso, arms, decorative wheels, hidden base, face aperture, badge cone, calyx socket	Master reference component
02_Faceplate_Black	Black	Face panel, ear cups, headband, microphone boom; recessed eye and mouth openings	Four locating pins + perimeter shoulder
03_Logo_Badge_White	White	Oval, slightly raised emblem base	Two locating pins + shallow chest recess
04_Calyx_Green	Morning Star green	Stem and simplified thick leaves	Keyed peg + shoulder stop

Suggested CAD file outputs

Native parametric assembly, STEP AP214/AP242 assembly, one 3MF per part, one assembled 3MF for viewing, and STL exports only as final printer interchange files.

2. Nominal Dimensions

Feature	Nominal	Engineering intent
Overall assembled height	100 mm	Includes calyx
Overall width	72 mm	Wheel-to-wheel / widest point
Overall depth	60 mm	Front face to rear shell
Body height without calyx	92 mm	Red body assembly
Nominal shell wall	2.5 mm	Use 2.5-3.0 mm around large cavities
Faceplate visible width	58 mm	Rounded rectangle
Faceplate visible height	36 mm	Rounded rectangle
Logo badge width	40 mm	White oval insert
Logo badge height	24 mm	White oval insert
Calyx overall width	52 mm	Leaf tip to leaf tip
Calyx overall height	34 mm	Stem plus leaf crown
Eye recess depth	0.8-1.0 mm	For white paint marker or UV resin fill

Proportion targets

Keep the tomato head slightly wider than tall and mildly flattened, following the supplied Morning Star tomato silhouette rather than a perfect sphere. The face should dominate the front view. The hidden base must carry the figure's weight; wheels are visual elements and should sit approximately 0.3-0.6 mm above the desk plane.

3. Joint Engineering

Faceplate joint

Use a continuous perimeter shoulder to establish flush depth, plus four tapered pins to establish position. Make the geometry do the alignment; CA glue should only retain the part. Add a 0.3-0.5 mm lead-in chamfer and hidden glue reservoirs behind the contact surface.

Logo badge joint

Use two pins so the badge cannot rotate. The chest recess should be 1.5 mm deep while the badge is approximately 2.0 mm thick, leaving it slightly proud. Add a small perimeter gap or shadow line of about 0.25 mm for a deliberate automotive-emblem appearance.

Calyx joint

Use a keyed connector, not a round peg. A cross, D-shape, or double-flat peg is acceptable. Include a shoulder that bottoms against the head so insertion depth is controlled. Keep the leaves thick and smoothly blended for reliable FDM printing.

CA glue control

Provide glue relief pockets 0.4 mm deep behind all cosmetic joints. Avoid applying CA to the visible perimeter. Dry-fit first, then place one or two micro-drops inside the pockets using a fine applicator. Thin CA can fog nearby surfaces; gel or medium CA is easier to control.

4. Tolerance Validation

Do not lock fit clearances from a generic number alone. Printer, nozzle, material, drying, and slicer compensation all matter. Print the included OpenSCAD fit coupon using the exact printer, nozzle, filament type, and profile planned for Mater.

Joint	Starting clearance per side	Adjustment rule
Faceplate pins/pockets	0.15-0.25 mm	Increase by 0.05 mm if force is excessive; reduce if part rattles
Logo pins/holes	0.15-0.25 mm	Target firm hand insertion without tools
Calyx keyed peg/socket	0.20-0.30 mm	Top-facing sockets often print slightly tighter
Visible perimeter shadow gap	0.20-0.30 mm	Keep consistent around faceplate and badge

Prototype acceptance

A dry assembly should be possible by hand, hold together during gentle handling, sit flat without rocking, and show even gaps around all inserts. No part should require pliers, hammering, or sanding to seat.

5. Printability Constraints

Constraint	Target
Primary process	FDM/FFF, PLA prototype
Nominal nozzle	0.4 mm
Layer height	0.20 mm prototype; 0.12-0.16 mm presentation print
Minimum cosmetic wall	2.5 mm
Minimum isolated feature	1.2 mm
Overhang target	45 degrees or less where practical
Support strategy	Avoid supports on visible face/head surfaces
Body orientation	Upright on hidden flat base
Faceplate orientation	Visible surface up or tilted to preserve finish
Badge orientation	Flat back on bed, logo face upward
Calyx orientation	Stem/connector oriented to minimize leaf supports

Logo production note

Use the supplied tomato image only as a visual reference. Obtain the original vector logo from Morning Star for the badge artwork. For a no-paint result, the white badge may be printed separately and decorated by a color printer, UV print, waterslide decal, or multicolor filament system. A single-color white badge cannot reproduce the full red/green logo by itself.

6. Prototype Test Plan

Stage	Action	Pass condition
Coupon	Print peg/hole and keyed-fit coupon	Select repeatable hand-fit clearance
Prototype A	Print all four parts at 100 mm	All parts seat; no interference
Dry fit	Assemble without glue	Even seams; stable; no visible distortion
Glue test	Use CA on one sacrificial or first article	No squeeze-out or frosting
Desk stability	Place on hard flat surface	No rocking; wheels appear to touch visually
Visual review	Front/side/back photographs	Matches frozen character proportions
Release	Export STEP, 3MF and STL	Files named, versioned and archived

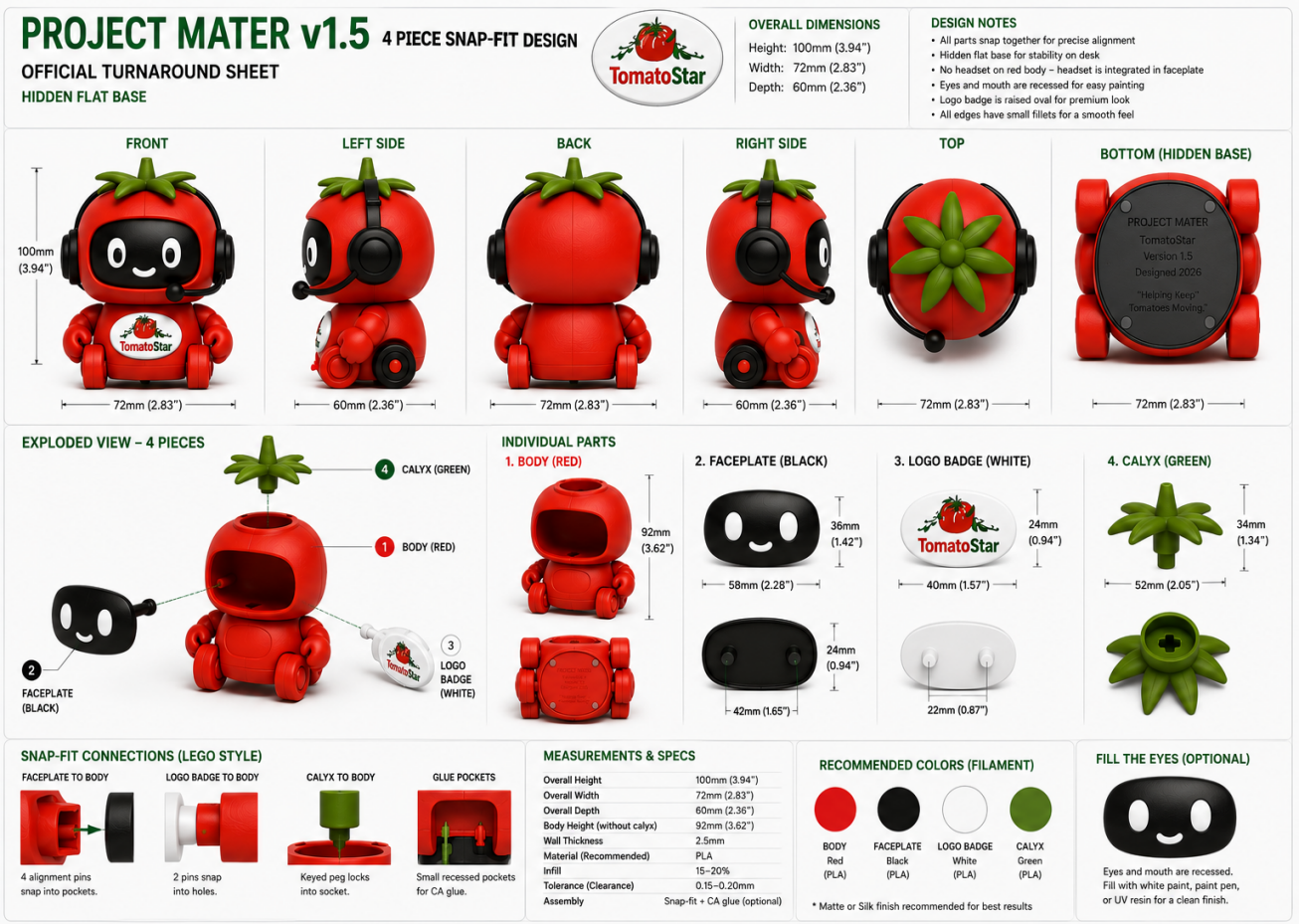
Suggested revision naming

Mater_v1_5_A (first printable), Mater_v1_5_B (fit correction), Mater_v1_5_RC1 (release candidate), Mater_v1_5_R1 (approved release). Record printer, filament, nozzle, slicer profile, and selected clearance with every prototype.

7. Source References



Morning Star tomato reference supplied by Robby. Match the calyx green to the approved Morning Star brand artwork rather than sampling from AI renders.



Concept turnaround. Use for appearance and component intent only; resolve inconsistencies in favor of the written engineering requirements.